

GROUP NUMBER:23

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AVAILABLE INPUTS: Input files are available in the hdfs file system: /data/BDC2526

TEST: The goal of this test is **to assess the scalability of your MRFairFFT program**. The test must be performed on the artificial datasets ArtificialNx DyKz.csv found on the hdfs file system. Conventionally, the name of the dataset represents some basic information: ArtificialNx DyKz.txt contains x points in R^y forming z natural clusters.

Use this table to report the results of the requested experiments. For all experiments use $kA=8$, $kB=2$, $L=16$, and filling only those cells for which the execution required at most 5 minutes.

Dataset	Runtime (2 executors)	Runtime (4 executors)	Runtime (8 executors)	Runtime (16 executors)	Obj. function (16 executors)
ArtificialN130kD7K10	3304 ms	1691 ms	903 ms	492 ms	38.4067580783 474
ArtificialN1.3mD7K10	27347 ms	14191 ms	7029 ms	3585 ms	39.2372376462 4849
ArtificialN13mD7K10	266896 ms	135017 ms	69364 ms	35146 ms	39.5674163303 6578
ArtificialN130mD7K10	Out Of Memory	Out Of Memory	Out Of Memory	506262 ms (128 executors and $L=128$)	39.6645875521 8468

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? YES
- (Only if you answered YES to previous question) Describe in at most 10 lines how you used the Generative AI tool.
Artificial intelligence was used only during an initial and limited exploratory phase of the project. The first two functionalities (points 1 and 2 of the assignment) were fully designed and implemented independently by the two group members without any use of AI tools. While waiting to proceed jointly with the development of point 3, an AI-generated class was temporarily used solely to test the manually developed functionalities. Before starting the implementation of point 3, all AI-generated testing classes were removed, as they were not required or included in the final submission